NumPy

NumPy, *short for Numerical Python, is a powerful library in Python used for numerical computing. NumPy is designed to handle large datasets effectively and efficiently, making it a fundamental tool in* ***scientific computing and data analysis.***

Key features of NumPy include:

- ***Arrays:***
- ***Integration with other libraries:***
- ***Performance:***
- ***Mathematical functions:***

Arrays: *NumPy's primary object is the ndarray (N-dimensional array), a flexible data structure that allows for efficient computation on large datasets.*

Integration with other libraries: *NumPy is often used alongside other libraries in the scientific Python ecosystem, such as*

- ***SciPy (for scientific computing),***
- ***Matplotlib , Seaborn, (for plotting),***
- ***sklearn (for AI)***
- ***pandas (for data analysis)***

Performance: *NumPy is implemented in C(programming Language) and Fortran, which ensures high performance for numerical computations. Additionally, NumPy arrays are more memory-efficient compared to Python lists.*

Mathematical functions: *NumPy provides a wide range of mathematical functions that operate element-wise on arrays, including trigonometric, statistical, and linear algebra functions. Overall, NumPy is a fundamental library for numerical computations in Python and serves as the foundation for many other libraries and applications in scientific computing and data analysis.*

Example of Mathematical functions:

Basic Mathematical Operations:

numpy.add(): *Add arguments element-wise.*
numpy.subtract(): *Subtract elements of the second array from the first array element-wise.*
numpy.multiply(): *Multiply arguments element-wise.*
numpy.divide(): *Divide the first input array by the second input array element-wise.*
numpy.floor_divide(): *Return the largest integer smaller or equal to the division of the inputs.*
numpy.power(): *First array elements raised to powers from the second array, element-wise.*
numpy.mod(): *Return the element-wise remainder of the division.*
numpy.abs(): *Calculate the absolute value element-wise.*
numpy.square(): *Return the element-wise square of the input.*
numpy.sqrt(): *Return the non-negative square-root of an array, element-wise.*
numpy.exp(): *Calculate the exponential of all elements in the input array.*
numpy.log(): *Natural logarithm, element-wise.*

Trigonometric Functions:

numpy.sin(), numpy.cos(), numpy.tan(): *Compute sine, cosine, and tangent element-wise.*
numpy.arcsin(), numpy.arccos(), numpy.arctan(): *Compute inverse sine, cosine, and tangent element-wise.*
numpy.deg2rad(): *Convert angles from degrees to radians.*
numpy.rad2deg(): *Convert angles from radians to degrees.*

Hyperbolic Functions:

numpy.sinh(), numpy.cosh(), numpy.tanh(): *Compute hyperbolic sine, cosine, and tangent element-wise.*
numpy.arcsinh(), numpy.arccosh(), numpy.arctanh(): *Compute inverse hyperbolic sine, cosine, and tangent element-wise.*

Rounding and Remainder Functions:

numpy.ceil(): *Return the ceiling of the input, element-wise.*
numpy.floor(): *Return the floor of the input, element-wise.*
numpy.round(): *Round elements of the array to the nearest integer.*
numpy.trunc(): *Return the truncated value of the input, element-wise.*

Statistics Functions:

numpy.mean(): *Compute the arithmetic mean along the specified axis.*
numpy.median(): *Compute the median along the specified axis.*
numpy.std(): *Compute the standard deviation along the specified axis.*
numpy.var(): *Compute the variance along the specified axis.*

These are just a few examples of mathematical functions available in NumPy. The library provides many more functions for various mathematical operations. You can find the complete list in the